

Bounds on Bipartite Ramsey Numbers

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NYC Discrete Math REU

August 2nd, 2021

Multicolor Ramsey Numbers

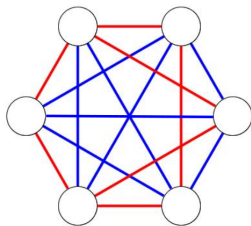
Definition (Multicolor Ramsey Numbers).

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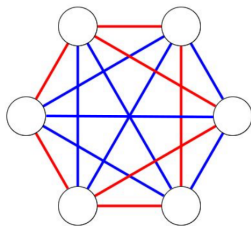


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Longstanding bounds

$$2^{t\ell/4} < r(t; \ell) < \ell^{t\ell}.$$

Recent improvements

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Sawin (2021).

$$r(t; \ell) > e^{0.266027(\ell-2)t} 2^{\frac{t}{2}+o(t)} \approx 1.305^{t\ell}.$$

Main ideas used for these improvements

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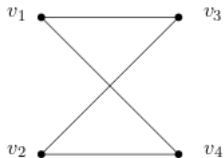
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Let's see how this sampling is done non-bipartite setting.

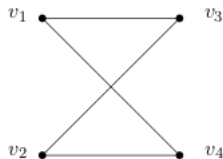
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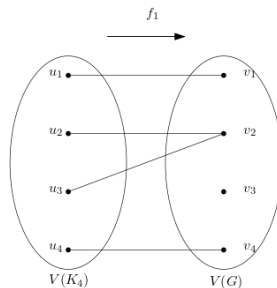


Graph G to be sampled from

Example of sampling method for 3-coloring a K_4



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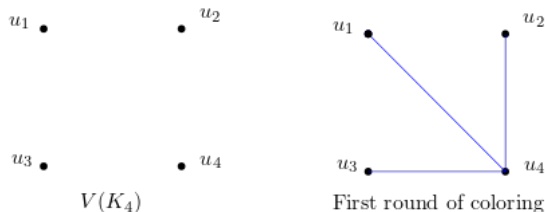
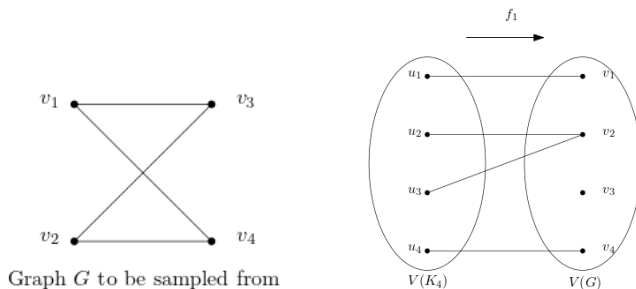


u_1 • u_2

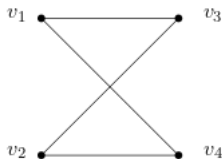
u_3 • u_4

$V(K_4)$

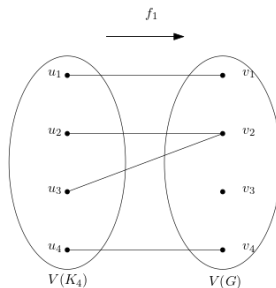
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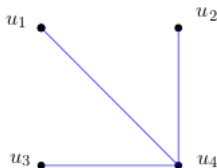
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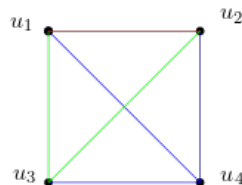
Graph G to be sampled from



$V(K_4)$



First round of coloring



Last two colors

Coloring K_N

Random functions for first $\ell - 2$ colors

Choose $\ell - 2$ independent uniformly random functions

$f_1, \dots, f_{\ell-2} : V(K_N) \rightarrow V(G)$. We color an edge $\{u, v\} \in E(K_N)$ by the i th color if its image $\{f_i(u), f_i(v)\} \in E(G)$. If there are collisions (i.e., $\{f_i(u), f_i(v)\} \in E(G)$ and $\{f_j(u), f_j(v)\} \in E(G)$), then we color $\{u, v\}$ by the minimum between i and j .

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Final two colors

For each pair of vertices $u, v \in V(K_N)$ that had no $f_i(u)$ and $f_i(v)$ connected by an edge G , we color them independently with probability $1/2$ either $\ell - 1$ st or ℓ th color.

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Notation

Let G be a graph with no K_t . We denote the probability of t independent uniformly random chosen vertices forming an independent set in G as $\mathbb{P}_G(t, t)$. Let $\mathbb{P}(t, t)$ denote the infimum of this probability over all such G .

Bounds given by this process

Lemma 1

For any $\ell, t \geq 3$,

$$r(t; \ell) > \mathbb{P}(t, t)^{-\frac{\ell-2}{t}} 2^{\frac{t-1}{2}}.$$

Sketch of a proof for lemma 1

- Let G be a graph with no K_t and probability at most $\mathbb{P}(t, t)$ that t random vertices from G form independent set.

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- Since G has no K_t , our random functions will not form a monochromatic K_t of the first $\ell - 2$ colors in K_N .
- So, to get a bound for $r(t; \ell)$ it suffices to show that the probability of forming a K_t in the last two colors of K_N is less than 1.

Adapting sampling method for $b(t; \ell)$

Adapted random functions

Consider bipartite graphs $K_{N,N}$ with vertex sets U and V and $G = (X, Y, E)$. Choose independent uniformly random injective functions $f_1, \dots, f_{\ell-2} : U \rightarrow X$ and $g_1, \dots, g_{\ell-2} : V \rightarrow Y$. We color an edge $\{u, v\} \in E(K_{N,N})$ by the i th color if its image $\{f_i(u), g_i(v)\} \in E(G)$. If there are collisions, then we color $\{u, v\}$ by minimum i . Apply last two colors in the same way as before.

Adapted $\mathbb{P}(t, t)$

Consider the bipartite graph $G = (X, Y, E)$ with no $K_{t,t}$. Let $\mathbb{P}'_G(t, t)$ denote the probability that t distinct independent uniformly random chosen vertices from X and t distinct independent uniformly random from Y form a $\overline{K_{t,t}}$.

Future work

- Fully adapt lemma, and proceed to search for a magic graph to apply it to.

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- We are currently trying to adapt Conlon and Ferber's algebraic construction.

Thanks!

I am probably out of time at this point, but thank you to everyone in the REU for making this a great experience.